Chris Lee

Ph.D. Candidate in Operations Research

Department of Industrial and Operations Engineering
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Research Interests

Online learning and sequential decision-making under structured feedback; bandit algorithms; Bayesian models of sequential decisions; applications in workforce management, energy, and healthcare.

Education

University of Michigan **2024–2029 (expected)**
Ph.D. in Operations Research
Advisors: Xiuli Chao and Izak Duenyas.

University of Washington **2018–2022**
B.S. in Mathematics with Honors
Honors thesis: *Computing Mixing Times of Parallel Tempered Markov Chains.* Advisors: Soumik Pal and Rekha Thomas.

Research Papers

Published

1. Nick Chiem, William Dudarov, **Chris Lee**, Sean Lee, and Kevin Liu. "A Characterization of Graphs with at Most Four Boundary Vertices." *Journal of Combinatorics*, 15(3), 2024. arXiv:2209.04438. *Authors listed alphabetically.*

Under Review

1. **Chris Lee**, Xiuli Chao, and Izak Duenyas. "Sequential Hiring of Contingent Workers Through Learning-Based Optimization." *Major Revision at Operations Research*, 2026.

In Preparation

1. **Chris Lee**, Xiuli Chao, and Izak Duenyas. "The Price of Shared Shocks in Combinatorial Semi-Bandits." *Working paper*, 2026.
2. **Chris Lee**, Xiuli Chao, and Izak Duenyas. "Dispatch of a Demand Response Virtual Power Plant via Online Learning" *Working paper*, 2026.

Research Positions

Washington Experimental Mathematics Lab **2021–2022**
Department of Mathematics, University of Washington
Conducted research on graph boundary structure, Markov-chain mixing, graphical designs, and spectral graph theory through an honors thesis, the Washington Experimental Mathematics Lab, and the Washington Directed Reading Program. Faculty mentors: Stefan Steinerberger, Rekha Thomas, and Soumik Pal; graduate mentors: Kevin Liu and Catherine Babecki.

Teaching Experience

Linear Programming (Graduate Level) **2026**
Teaching Assistant, University of Michigan

Project Management **2025, 2026**
Teaching Assistant, University of Michigan

Advanced Linear Algebra Tools and Applications **2021**
Undergraduate Grader, University of Washington

Industry Experience

Lead Data Analyst **2022–2024**
AmFam Group (American Family Insurance)
Led the architecture and development of a year-long initiative to automate a central actuarial ratemaking workflow over terabyte-scale data, working on a five-person team.